4.1.2. Transamination (AST and ALT)

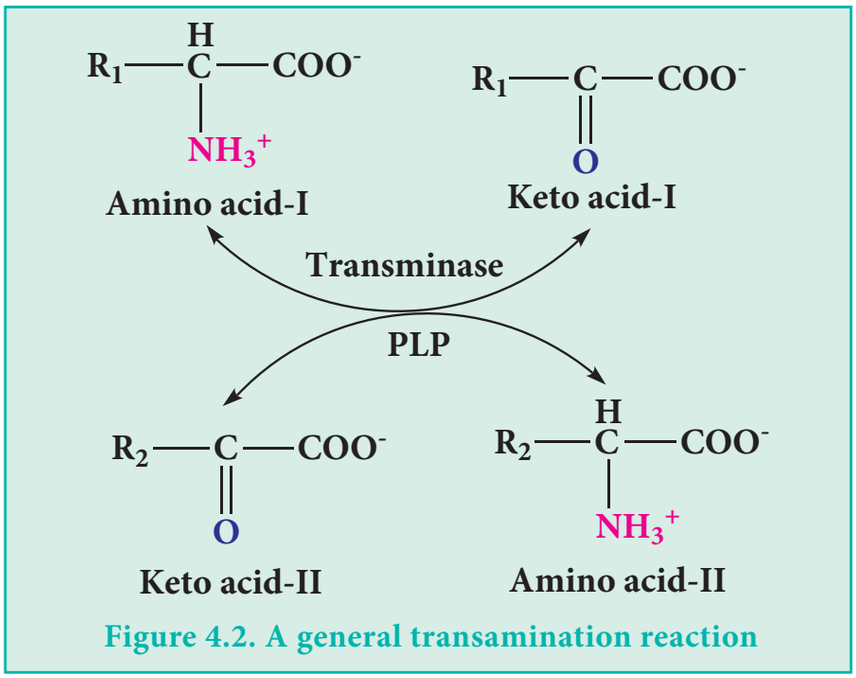

Figure 4.2. A general transamination reaction

Transamination is a process, in which transfer of amino groups occur between a keto acid and an amino acid. It is just shuffling of amino groups and not removal of amino groups. Each amino acid has a specific transaminase (Figure 4.2). However, Alanine transaminase (ALT) and Aspartate transaminase (AST) are the important transaminases.

The reactions catalysed by ALT and AST are :

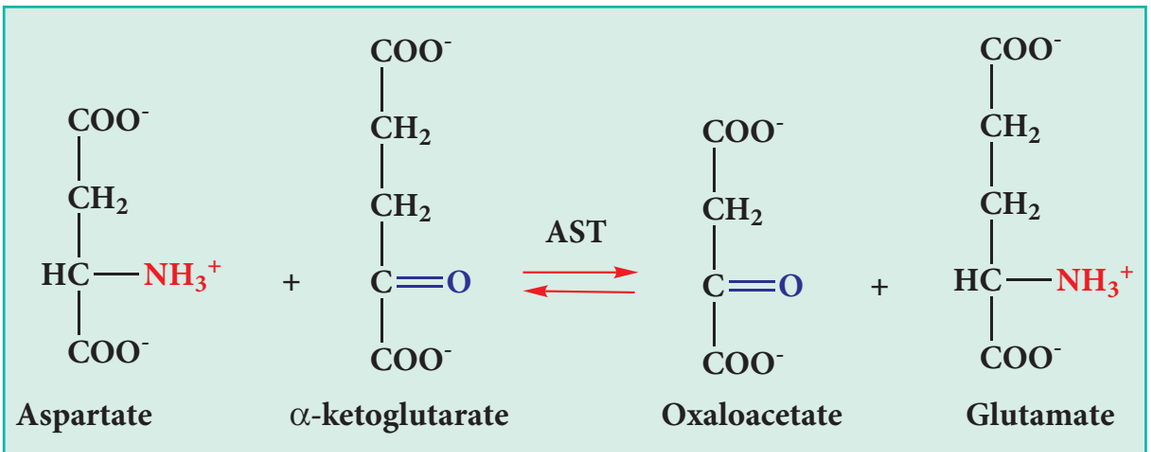

Fig 4.3. Reaction catalysed by aspartate amino transferase

Aspartate transaminase catalyses the inter-conversion of aspartate and α-ketoglutarate to oxaloacetate and glutamate (Figure 4.3), while alanine transaminase catalyses the conversion of alanine and α- ketoglutarate to pyruvate and glutamate (Figure 4.4).

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